

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Enterprise Java

Practical-6

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Class:TYIT	Batch:1
Date of Assignment: 22-8-2026	Date/Time of Submission: 22-8-2026

6. Implement the following Intermediate JSP Applications.

a. Develop a JSP application to demonstrate the use of JSP Directives and Implicit Objects.

Code:

```
<%@ page language="java" contentType="text/html;
charset=UTF-8" %>
<%@ page import="java.util.Date" %>
<html>
  <head>
    <title>JSP Directives and Implicit Objects</title>
  </head>
  <body>
    <h2>JSP Directives and Implicit Objects</h2>
    <h3>JSP Directives</h3>
    <p>Language: Java</p>
    <p>Content Type: <%=
response.getContentType()%></p>
    <p>Current Date: <%= new Date()%></p>
    <h3>Implicit Objects</h3>
    <p>1. Request Method: <%=
request.getMethod()%></p>
    <p>2. Response Content Type: <%=
response.getContentType()%></p>
    <p>3. Session ID: <%= session.getId()%></p>
    <p>4. Application Server: <%=
application.getServerInfo()%></p>
    <p>5. Page Name: <%=
page.getClass().getName()%></p>
    <p>6. PageContext: <%=
pageContext.getClass().getName()%></p>
    <p>7. Request Parameter: <%=
request.getParameter("name")%></p>
    <p>8. Config Servlet Name: <%=
config.getServletName()%></p>
```

```
<p>9. Exception Object: Exception object is available
only in error pages.</p>
```

```
<h2>radhika 17</h2>
</body>
</html>
```

Output:

JSP Directives and Implicit Objects

JSP Directives

Language: Java

Content Type: text/html; charset=UTF-8

Current Date: Sat Aug 22 11:28:30 IST 2026

Implicit Objects

1. Request Method: GET

2. Response Content Type: text/html; charset=UTF-8

3. Session ID: 80c0d86df3b7661827f3391e869c

4. Application Server: GlassFish Server Open Source Edition 4.1.1

5. Page Name: org.apache.jsp.index_jsp

6. PageContext: org.apache.jasper.runtime.PageContextImpl

7. Request Parameter: null

8. Config Servlet Name: jsp

9. Exception Object: Exception object is available only in error pages.

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b. Develop a JSP application to demonstrate Form Processing and Input Validation.

Code:

Index.jsp:

```
<html>
<head>
<title>Patient Registration Form</title>
</head>
<body>
<h2>Patient Registration Form</h2>
<form action="processPatient.jsp" method="post">
  Patient Name:
  <input type="text" name="name">
  <br><br>
  Age:
  <input type="text" name="age">
  <br><br>
  Email:
  <input type="text" name="email">
  <br><br>
  Mobile:
  <input type="text" name="mobile">
  <br><br>

  Gender:
  <input type="radio" name="gender" value="Male">
Male
  <input type="radio" name="gender"
value="Female"> Female
  <br><br>

  Department:
  <select name="department">
    <option value="">Select Department</option>
    <option value="General Medicine">General
Medicine</option>
    <option
value="Cardiology">Cardiology</option>
    <option
value="Orthopedics">Orthopedics</option>
    <option
value="Dermatology">Dermatology</option>
  </select>
  <br><br>

  Symptoms:
  <input type="checkbox" name="symptom"
value="Fever"> Fever
  <input type="checkbox" name="symptom"
value="Cough"> Cough
  <input type="checkbox" name="symptom"
value="Headache"> Headache
  <br><br>

  Address:
  <textarea name="address"></textarea>
  <br><br>
  <input type="submit" value="Register Patient">
</form>
</body>
</html>
```

processPatient.jsp:

```
<html>
<head>
<title>Patient Details</title>
</head>
<body>
<h2>Patient Details</h2>
<%
  String name = request.getParameter("name");
  String age = request.getParameter("age");
  String email = request.getParameter("email");
  String mobile = request.getParameter("mobile");
  String gender = request.getParameter("gender");
  String department =
request.getParameter("department");
  String address = request.getParameter("address");
  String[] symptoms =
request.getParameterValues("symptom");
  boolean valid = true;
  if (name == null || name.trim().equals("")) {
    out.println("Patient name is required.<br>");
    valid = false; }
  if (age == null || age.trim().equals("")) {
    out.println("Age is required.<br>");
    valid = false; }
  if (email == null || email.trim().equals("")) {
    out.println("Email is required.<br>");
    valid = false; }
  if (mobile == null || mobile.trim().equals("")) {
    out.println("Mobile number is required.<br>");
    valid = false; }
  if (gender == null) {
    out.println("Please select gender.<br>");
    valid = false; }
  if (department == null || department.equals("")) {
    out.println("Please select a department.<br>");
    valid = false; }
  if (symptoms == null) {
    out.println("Please select at least one
symptom.<br>");
    valid = false; }
  if (address == null || address.trim().equals("")) {
    out.println("Address is required.<br>");
    valid = false; }
  if (valid) {
%>
<h3>Patient Registration Successful</h3>
Patient Name: <%= name%><br><br>
Age: <%= age%><br><br>
Email: <%= email%><br><br>
Mobile: <%= mobile%><br><br>
Gender: <%= gender%><br><br>
Department: <%= department%><br><br>
Symptoms:
<%
  for (String symptom : symptoms) {
    out.println(symptom + " ");
  }
```

```

%>
<br><br>
Address: <%= address%><br><br>
<%
}

```

```

%>
</body>
</html>

```

Output:

Patient Registration Form

Patient Name:

Age:

Email:

Mobile:

Gender: ☐ Male ☒ Female

Department:

Symptoms: ☒ Fever ☐ Cough ☐ Headache

Address:

Patient Details

Patient Registration Successful

Patient Name: radhika

Age: 18

Email: radhmalimvlu@gmail.com

Mobile: 01234567890

Gender: Female

Department: General Medicine

Symptoms: Fever

Address: andheri e

c. Develop a JSP application to display data from a MySQL Database using JDBC.

Code:**Sql code:**

```

CREATE DATABASE ecommerce;
USE ecommerce;
Database changed
CREATE TABLE product (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(100),
    category VARCHAR(50),
    brand VARCHAR(50),
    price DECIMAL(10,2),
    quantity INT,
    rating DECIMAL(2,1),
    discount DECIMAL(5,2),
    description VARCHAR(255)
);

```

Index.jsp:

```

<%@ page language="java"
    contentType="text/html; charset=UTF-8"
    import="java.sql.*" %>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>E-Commerce Products</title>
<style>
    body {
        font-family: Arial, sans-serif;
        background-color: #f2f2f2;
        margin: 30px;
    }
    h2 {
        text-align: center;
        color: #333;
    }
    table {

```

```

width: 100%;
border-collapse: collapse;
background-color: white;
    }

    th {
        background-color: #2874A6;
        color: white;
        padding: 12px;
    }
    td {
        padding: 10px;
        border: 1px solid #ddd;
        text-align: center;
    }
    tr:nth-child(even) {
        background-color: #f2f2f2;
    }
    tr:hover {
        background-color: #D6EAF8;
    }
</style>
</head>
<body>
<h2>E-Commerce Product List</h2>
<table>
<tr>
<th>Product ID</th>
<th>Product Name</th>
<th>Category</th>
<th>Brand</th>
<th>Price</th>
<th>Quantity</th>
<th>Rating</th>
<th>Discount</th>
<th>Description</th>
</tr>
<%
    Connection con = null;

```

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```
Statement stmt = null;
ResultSet rs = null;
try {
    Class.forName("com.mysql.jdbc.Driver");
    con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/ecommerce",
        "root",
        "root"
    );
    stmt = con.createStatement();
    rs = stmt.executeQuery("SELECT * FROM
product");
    while (rs.next()) {
        <tr>
            <td><%= rs.getInt("product_id")%></td>
            <td><%= rs.getString("product_name")%></td>
            <td><%= rs.getString("category")%></td>
            <td><%= rs.getString("brand")%></td>
            <td><%= rs.getDouble("price")%></td>
            <td><%= rs.getInt("quantity")%></td>
            <td><%=
rs.getDouble("rating")%> ★</td>
        </tr>
    }
} catch (Exception e) {
    out.println("<p>Error: " + e.getMessage() +
"</p>");
} finally {
    try {
        if (rs != null) {
            rs.close();
        }
        if (stmt != null) {
            stmt.close();
        }
        if (con != null) {
            con.close();
        }
    } catch (Exception e) {
        e.printStackTrace();
    }
}
```

Output:

product_id	product_name	category	brand	price	quantity	rating	discount	description
1	iPhone 15	Mobile	Apple	69999.00	10	4.8	10.00	Apple iPhone with advanced camera and performance
2	Galaxy S24	Mobile	Samsung	64999.00	15	4.7	12.00	Samsung smartphone with AMOLED display
3	MacBook Air M2	Laptop	Apple	89999.00	8	4.9	8.00	Lightweight laptop powered by Apple M2 chip
4	WH-1000XM5	Headphones	Sony	29999.00	20	4.6	15.00	Wireless noise cancelling headphones
5	Inspiron 15	Laptop	Dell	55999.00	12	4.5	10.00	Dell laptop suitable for everyday computing

5 rows in set (0.00 sec)

E-Commerce Product List								
Product ID	Product Name	Category	Brand	Price	Quantity	Rating	Discount	Description
1	iPhone 15	Mobile	Apple	₹69999.0	10	4.8 ⭐	10.0%	Apple iPhone with advanced camera and performance
2	Galaxy S24	Mobile	Samsung	₹64999.0	15	4.7 ⭐	12.0%	Samsung smartphone with AMOLED display
3	MacBook Air M2	Laptop	Apple	₹89999.0	8	4.9 ⭐	8.0%	Lightweight laptop powered by Apple M2 chip
4	WH-1000XM5	Headphones	Sony	₹29999.0	20	4.6 ⭐	15.0%	Wireless noise cancelling headphones
5	Inspiron 15	Laptop	Dell	₹55999.0	12	4.5 ⭐	10.0%	Dell laptop suitable for everyday computing